Skylar: Troubleshooting Metadata Export

Guide	Skylar
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IMPORTANT UPDATES FOR 12.3.7+

In 12.3.7, 12.5.1, and 12.6.0+, the metadata exporter has undergone a significant change to the way it operates. For more info on specific troubleshooting, please reach out to #skylar-data-team on Slack as this document will be updated as new issues may be found.

1) TENANT: Process and Log Information

Metadata exporter runs as an SL1 process called `Enterprise Database: Skylar Metadata Exporter`.

Logs are located at: `/var/log/sl1/run_skylar_metadata_exporter.log`. It may be helpful to turn debug logs on by switching `Debug Mode` to enabled for the `Enterprise Database: Skylar Metadata Exporter` process in the UI, and then gather logs after another metadata export run has been completed.

SL1 < 12.3.7: By default it runs every hour, so waiting 1-2 hours should be enough time.

SL1 >= 12.3.7: The metadata exporter process runs every 1 minute, with snapshots still being sent hourly.

Alternatively, after enabling debug mode on the process, you could switch the `Operating State` to Disabled and then run the exporter manually:

```
 nohup sudo -u s-em7-core
/opt/em7/backend/run_skylar_metadata_exporter.py --run-snapshot >
/dev/null 2>&1 &
```

Check the log file to ensure the run has completed (or failed).

> ⚠ If running manually don't forget to re-enable the process for automated scheduling.

2) (SL1 < 12.3.7) TENANT: Exporter is not completing in time allotted

There is likely a lot of metadata in this case. Increase the run time on the SL1 process to allow for more time to complete.

3) CLUSTER: "Error communicating with the metadata-snapshot-service" in logs

Check that the Skylar endpoint is correct.

Check the `metadata-snapshot-service` pod health and logs in Skylar.

```
kubectl logs <metadata-snapshot-service-pod>
```

```
kubectl describe pod <metadata-snapshot-service-pod>
```

4) CLUSTER: Unknown database - exporter aborting snapshot

Ticket reference that will fix this issue: [EM-72925]

See the linked ticket for full details, and for what SL1 version the fix is expected, but here's the relevant log that was being hit:

```
2025-03-14T19:17:04.911697+00:00 localhost
silo.run_skylar_metadata_exporter.271[913980]: ERROR service.run 150
Error during non-HTTP functionality. Likely indicates a bug. Aborting
snapshot build and tearing down.#012Traceback (most recent call
last):#012  File "/opt/em7/services/skylar-metadata-
exporter/skylar_metadata_exporter/service.py", line 141, in run#012
_build_snapshot(dbc, logger)#012  File "/opt/em7/services/skylar-
metadata-exporter/skylar_metadata_exporter/service.py", line 111, in
_build_snapshot#012    gather.gather(dbc, send_func=send_batch,
max_rows=config.BATCH_SIZE, logger=logger)#012  File
"/opt/em7/services/skylar-metadata-
exporter/skylar_metadata_exporter/gather.py", line 77, in gather#012
for batch in gathered_batches:#012  File "/opt/em7/services/skylar-
metadata-exporter/skylar_metadata_exporter/gather.py", line 36, in
__next__#012    return self.iter.__next__()#012
^^^^^^^^^^^^^^^^^^^^#012  File "/opt/em7/services/skylar-metadata-
exporter/skylar_metadata_exporter/sources/device_config.py", line 209,
in gather#012    config_data = _get_config_data(dbc, app_id, did,
logger)#012
^^^^^^^^^^^^^^^^^^^
```

5) TENANT: MemoryError

Some queries to the SL1 db can be larger (e.g., dynamic app config data for a device, index label table), and require more memory. Since the exporter is an SL1 managed process, it abides by an internally set memory limit of 1 GiB. Breakage of this limit kills the process.

Workaround is to increase the memory limit aligned to the metadata exporter process (add to `/etc/silo.conf`):

```
[PROC_VIRTUAL_MEM_LIMIT]
271 = 2147483648
```

Example traceback:

```
2025-05-07T13:27:24.009388+00:00 localhost
silo.run_skylar_metadata_exporter.271[1469643]: ERROR service.run 150
Error during non-HTTP functionality. Likely indicates a bug. Aborting
snapshot build and tearing down.#012Traceback (most recent call
last):#012  File "/opt/em7/services/skylar-metadata-
exporter/skylar_metadata_exporter/service.py", line 141, in run#012
_build_snapshot(dbc, logger)#012  File "/opt/em7/services/skylar-
metadata-exporter/skylar_metadata_exporter/service.py", line 111, in
_build_snapshot#012    gather.gather(dbc, send_func=send_batch,
max_rows=config.BATCH_SIZE, logger=logger)#012  File
"/opt/em7/services/skylar-metadata-
exporter/skylar_metadata_exporter/gather.py", line 77, in gather#012
for batch in gathered_batches:#012  File "/opt/em7/services/skylar-
metadata-exporter/skylar_metadata_exporter/gather.py", line 36, in
__next__#012    return self.iter.__next__()#012
^^^^^^^^^^^^^^^^^^^^#012  File "/opt/em7/services/skylar-metadata-
exporter/skylar_metadata_exporter/sources/perf_index_label.py", line
246, in gather#012    new_results = _gather_by_app_id(dbc, app_id,
max_rows)#012
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^#012  File
"/opt/em7/services/skylar-metadata-
exporter/skylar_metadata_exporter/sources/perf_index_label.py", line
152, in _gather_by_app_id#012    results = dbc.autofetchall_dict(#012
^^^^^^^^^^^^^^^^^^^^^^^^#012  File "/opt/em7/services/skylar-metadata-
exporter/sl_mysql/cursor.py", line 567, in autofetchall_dict#012
self.execute(query, args, log=log, logger=logger, **kwargs)#012  File
"/opt/em7/services/skylar-metadata-exporter/sl_mysql/cursor.py", line
120, in wrapper#012    return func(*vargs, **kwargs)#012
^^^^^^^^^^^^^^^^^^^^^^^#012  File "/opt/em7/services/skylar-metadata-
exporter/sl_mysql/cursor.py", line 99, in callFunc#012    result =
func(obj, *args, **kwargs)#012
^^^^^^^^^^^^^^^^^^^^^^^^^^^#012  File "/opt/em7/services/skylar-
metadata-exporter/sl_mysql/cursor.py", line 464, in execute#012    r =
super(LegacySiloCursor, self).execute(query, args)#012
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^#012  File
"/opt/em7/services/skylar-metadata-exporter/MySQLdb/cursors.py", line
206, in execute#012    res = self._query(query)#012
^^^^^^^^^^^^^^^^^^^#012  File "/opt/em7/services/skylar-metadata-
exporter/MySQLdb/cursors.py", line 321, in _query#012
self._post_get_result()#012  File "/opt/em7/services/skylar-metadata-
exporter/MySQLdb/cursors.py", line 355, in _post_get_result#012
self._rows = self._fetch_row(0)#012
^^^^^^^^^^^^^^^^^^#012  File "/opt/em7/services/skylar-metadata-
exporter/MySQLdb/cursors.py", line 328, in _fetch_row#012    return
self._result.fetch_row(size, self._fetch_type)#012
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^#012MemoryError
```

6) (SL1 >= 12.3.7) TENANT: RuntimeError

There is a slight increase in demand for resources by skylar metadata exporter with the addition of exporting cleared events and other data types to Skylar at a higher frequency through the same process. In the case of a memory error,

a runtime error will be seen:

Example traceback:

```
2025-07-22T13:36:04.572955+00:00 localhost silo.run_skylar_metadata_exporter.271[2234615]: DEBUG cleared_events.export_event_clears 80 error#012Traceback (most recent call last):#012  File "/opt/em7/services/skylar-metadata-exporter/skylar_metadata_exporter/cleared_events.py", line 77, in export_event_clears#012 _otel_logger_setup("sl.sl1.metadata.events.cleared", EXPORT_LOG_NAMESPACE)#012  File "/opt/em7/services/skylar-metadata-exporter/skylar_metadata_exporter/cleared_events.py", line 45, in _otel_logger_setup#012    processor = BatchLogRecordProcessor(OTLPLogExporter())#012 ^^^
```